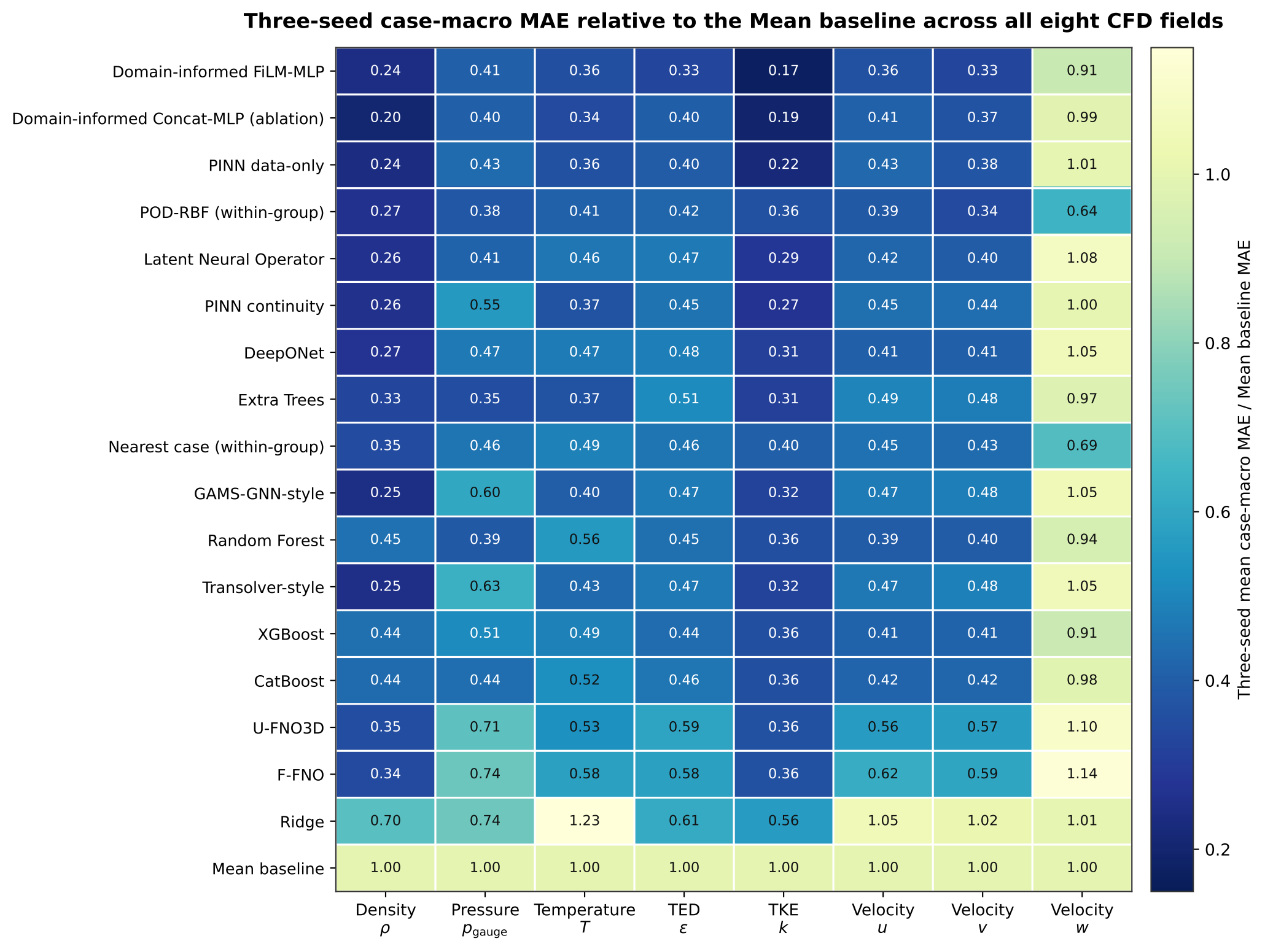
Three-seed case-macro MAE relative to the Mean baseline across all eight CFD fields
Domain-informed FiLM-MLP 0.24 0.41 0.36 0.33 0.17 0.36 0.33 0.91
Domain-informed Concat-MLP (ablation) 0.20 0.40 0.34 0.40 0.19 0.41 0.37 0.99
PINN data-only 0.24 0.43 0.36 0.40 0.22 0.43 0.38 1.01
POD-RBF (within-group) 0.27 0.38 0.41 0.42 0.36 0.39 0.34 0.64
Latent Neural Operator 0.26 0.41 0.46 0.47 0.29 0.42 0.40 1.08
PINN continuity 0.26 0.55 0.37 0.45 0.27 0.45 0.44 1.00
DeepONet 0.27 0.47 0.47 0.48 0.31 0.41 0.41 1.05
Extra Trees 0.33 0.35 0.37 0.51 0.31 0.49 0.48 0.97
Nearest case (within-group) 0.35 0.46 0.49 0.46 0.40 0.45 0.43 0.69
GAMS-GNN-style 0.25 0.60 0.40 0.47 0.32 0.47 0.48 1.05
Random Forest 0.45 0.39 0.56 0.45 0.36 0.39 0.40 0.94
Transolver-style 0.25 0.63 0.43 0.47 0.32 0.47 0.48 1.05
XGBoost 0.44 0.51 0.49 0.44 0.36 0.41 0.41 0.91
CatBoost 0.44 0.44 0.52 0.46 0.36 0.42 0.42 0.98
U-FNO3D 0.35 0.71 0.53 0.59 0.36 0.56 0.57 1.10
F-FNO 0.34 0.74 0.58 0.58 0.36 0.62 0.59 1.14
Ridge 0.70 0.74 1.23 0.61 0.56 1.05 1.02 1.01
Mean baseline 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
Density ρ
Pressure p_gauge
Temperature T
TED ε
TKE k
Velocity u
Velocity v
Velocity w
Three-seed mean case-macro MAE / Mean baseline MAE
1.0
0.8
0.6
0.4
0.2